# 🥕 Plant Disguise Lab



Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 30 min + 1 week

## 🥔 Vegetable Market Detective

> **You will learn:** Many vegetables are modified roots, stems or leaves — nodes, buds and scale leaves give them away.

### YOU NEED

- Carrot, radish, sweet potato, potato, ginger, onion, colocasia (arbi), cabbage
- A knife (with an adult)
- A plate with damp cotton

> ⚠️ **Safety:** Use the knife only with an adult's help.

### STEPS

1. Look at each vegetable for nodes, buds ("eyes"), scale leaves and root hairs.
2. Cut the onion vertically and find the stem disc and fleshy scale leaves.
3. Put a potato piece with an "eye" on damp cotton for a week and watch it sprout.
4. Do the same with a carrot top.
5. Decide whether each is a modified root, stem or leaf, and record the proof.

### OBSERVATION TABLE

| Vegetable | Root / stem / leaf | My proof | Food stored for |
|---|---|---|---|
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### QUESTIONS

1. Why is potato a stem and sweet potato a root?

2. Which part of the onion do we eat?

3. What does a modified stem have that a root never has?

### WHAT IS HAPPENING? — THE SCIENCE

Stems have nodes and buds; roots do not. The potato "eyes" are nodes with buds, so potato is a stem tuber; ginger is a rhizome; onion is a bulb with fleshy leaves on a short stem; carrot, radish and sweet potato store food in roots.

**Key terms:** tuber · rhizome · bulb · tap root · node

Activity 2 · Hands-on · 🧪 In the lab · ⏱ 45 minutes

## 🌺 Flower Dissection & Floral Formula

> **You will learn:** A flower has four whorls; the ovary shows how ovules are attached (placentation).

### YOU NEED

- A hibiscus (china rose) flower
- A tomato and a lemon
- A blade (with teacher help)
- White paper, glue
- Phone microscope

> ⚠️ **Safety:** Handle the blade carefully, cutting away from your fingers.

### STEPS

1. Remove and count the parts of each whorl: epicalyx, calyx, corolla, stamens and pistil.
2. Note whether parts are free or fused. In hibiscus the stamens are fused into a tube (monadelphous).
3. Stick each whorl on paper in order from outside to inside.
4. Cut the ovary across and look at it with the phone microscope.
5. Cut a tomato and a lemon across and compare how the seeds sit.
6. Write the floral formula of hibiscus.

### OBSERVATION TABLE

| Whorl | Number of parts | Free or fused | Colour |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

### QUESTIONS

1. What is monadelphous condition?

2. What type of placentation did you see in tomato?

3. Name the four whorls of a flower.

### WHAT IS HAPPENING? — THE SCIENCE

Calyx and corolla protect and attract; androecium (stamens) and gynoecium (pistil) are reproductive. Tomato and lemon show axile placentation — ovules attached to a central axis in a multilocular ovary.

**Key terms:** **calyx** **corolla** **androecium** **gynoecium** **axile placentation**

Digital activity

## 💻 Modification Match

> Tap a plant part, then tap what it really is.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch5/2**

### MODIFIED ROOT · MODIFIED STEM · MODIFIED LEAF

| Modified root | Modified stem | Modified leaf |
|---|---|---|
| | | |

Carrot · Banyan prop roots · Rhizophora pneumatophores · Maize stilt roots · Potato · Ginger · Opuntia phylloclade · Citrus thorn · Cucumber tendril · Pea tendril · Cactus spines · Onion fleshy scales · Pitcher plant pitcher

> **You will learn:** Look for the clue: stems have nodes and buds, leaves arise at nodes, roots have neither.

### THINK & WRITE

1. Pea tendril and cucumber tendril do the same job. Why are they in different groups?

2. What are such organs called?